THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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ANSWER KEY & MARKING SCHEME SET A

Chapter 4: **Electricity: Magnetic and Heating Effects** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** Hans Christian Oersted	In 1820
2	**(a)** the number of turns increases	More turns means stronger magnet
3	**(a)** it can be switched on and off	Lifts and releases iron objects
4	**(c)** harmful particles from space	Acts as a shield
5	**(d)** heated by electric current	Heating effect
6	**(b)** it becomes hot	Safety first
7	**(d)** Luigi Galvani	Late 1700s
8	**(c)** positive terminal	LED works in one direction only
9	**(c)** moist paste	Not a liquid
10	**(a)** e-waste recycling	Prevents pollution

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Oersted (1). In 1820 (1).	1+1
12	A coil of wire that behaves like a magnet when current flows (1). Strengthened by an iron core (1).	1+1
13	Amount of current (1). Number of turns of the coil or iron core (1).	1+1
14	By reversing the direction of current (1). Poles of the coil then interchange (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Place a wire over a compass needle (1). Pass current through the wire (1). Needle deflects, showing a magnetic effect (1).	1+1+1
16	Wind insulated wire around an iron nail (1). Connect ends to a cell (1). It picks up pins or clips while current flows (1).	1+1+1
17	More current gives stronger field (1). More turns gives stronger magnet (1). Iron core makes it much stronger (1).	1+1+1
	OR: Electromagnet can be switched on and off (1½). Can lift and release iron objects when needed (1½).	1½+1½
18	Magnitude of current (1). Material, thickness and length of wire (1). Duration of current (1).	1+1+1
	OR: Nichrome becomes hotter (1). It has higher resistance (1). More electrical energy changes to heat (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Compass deflects near current wire (1). Current produces magnetic field (1). Coil on iron core forms electromagnet (1). Three factors: current, turns, iron core (2).	1+1+1+2
	OR: Strong electromagnet hung from crane (1). Current ON lifts iron objects (1). Current OFF releases (1). Can be controlled (1). Permanent magnet cannot be switched off (1).	1+1+1+1+1
20	Resistance changes electrical energy to heat (1). Factors: current, material, thickness, length, time (2). Applications: heater, iron, kettle, lamp (2).	1+2+2
	OR: Connect a nichrome wire to one cell and touch briefly (1). Connect to two cells (1). Wire is hotter with two cells (1). More current gives more heat (1). Do not touch for long (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET B

Chapter 4: **Electricity: Magnetic and Heating Effects** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** gets deflected	Current produces a magnetic field
2	**(a)** more cells are connected	More current
3	**(c)** iron	Magnetic material
4	**(b)** resistance of the conductor	Electrical energy converts to heat
5	**(a)** electric iron	Contains a heating element
6	**(c)** electrolyte	Liquid or paste
7	**(c)** Alessandro Volta	Metals and liquid give current
8	**(b)** zinc and copper	Two different metals are needed
9	**(d)** a single-use cell	Disposed after use
10	**(d)** compass	Needle deflects

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Lifts iron and steel objects in factories (1). Releases them when current is switched off (1).	1+1
12	A conductor gets heated when current passes through it (1). Because of resistance, electrical energy converts to heat (1).	1+1
13	It has high resistance (1). It gets hot without melting easily (1).	1+1
14	Electric heater (1). Electric iron or kettle (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Electric heater (1). Electric iron or kettle (1). Incandescent lamp (1).	1+1+1
16	Overheating can melt plastic parts (1½). It may cause fire (1½).	1½+1½
17	Two different metals dip in an electrolyte (1). Chemical reaction occurs (1). Current flows from positive terminal through the circuit to negative (1).	1+1+1
	OR: Galvani's frog leg twitched with two metals (1). Volta used saltwater paper instead of frog (1). Showed metals and liquid make electricity (1).	1+1+1
18	Insert copper wire and iron nail in each lemon (1). Join several lemons in series and connect an LED (1). LED glows; reverse connections if it does not (1).	1+1+1
	OR: Zinc–copper, aluminium–copper, iron–copper (any three) (2). Copper acts as positive (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Diagram (1). Zinc container negative (1). Carbon rod with cap positive (1). Paste electrolyte (1). Chemical reaction produces current and cell is single-use (1).	1+1+1+1+1
	OR: Galvani noticed frog's leg kicking with copper and iron (1½). He believed electricity came from the frog (1). Volta believed it came from metals (1). He used saltwater-soaked paper and still got current (1). Led to the first battery (½).	1½+1+1+1+½
20	Insert copper and iron into lemons (1). Join several lemons and connect LED (1). LED glows (1). Metals are electrodes, juice is electrolyte (1). Chemical reaction produces current (1).	1+1+1+1+1
	OR: Dry cell is single-use (1). Rechargeable can be reused (1). Rechargeable saves money over time (1). Both contain harmful materials (1). E-waste must be recycled (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.